Do Artificial Neural Networks Construct Language-Specific Geometries of Time?

A Computational Investigation of Linguistic Relativity, Script Chirality, Causal Metaphor Steering, and Connectomic Ground Truths

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Abstract

Cognitive linguistics has long argued that human conceptualizations of time are structured through linguistic spatial metaphors. In landmark research, Boroditsky (2001) showed that English speakers construct horizontal timelines, while Mandarin speakers frequently scaffold time vertically (*shàng / xià*). Furthermore, cross-cultural studies demonstrate that writing systems shape mental timelines, causing Arabic speakers to structure chronologies from Right to Left. In this research report, I test whether artificial neural networks autonomously construct these language-specific spatio-temporal geometries without explicit supervision, and contrast them against non-linguistic biological wetware. Across five empirical experiments spanning English GPT-2, Chinese GPT-2, Arabic GPT-2, and the 139,248-neuron whole-brain connectome of *Drosophila melanogaster* (analyzing 111 verified circadian pacemaker and celestial compass neurons), I demonstrate: (1) English GPT-2 exhibits an overwhelmingly horizontal timeline ($p < 0.0001$, ratio up to 9.17×), while Chinese GPT-2 exhibits prominent vertical alignment peaks (up to 2.58× vertical preference); (2) Grounded physical concepts converge across languages, whereas abstract temporal concepts exhibit topological warping under Gromov-Wasserstein analysis; (3) Layer-7 activation patching confirms that spatial vectors act as causal steering levers over temporal predictions ($\beta = -0.0268$ in Chinese; $\beta = +0.0090$ in English), mirroring human cognitive priming; (4) Arabic GPT-2 mirrors the horizontal timeline to Right-to-Left (future is left: $+0.0725$, right: -0.0725), matching script chirality across the y-axis; and (5) Evaluating the 111 verified circadian pacemaker neurons (s-LNv, l-LNv, LNd, DN1) and central complex E-PG compass neurons from the FlyWire whole-brain connectome reveals that while biological wetware runs on an exact periodic limit cycle ($C = 14.58$), human language models flatten time into an artificial linear ray ($C = 0.31$–0.46). All code and data are open-sourced in a standalone repository.

Keywords: Linguistic Relativity, Cognitive Metaphor, Conceptual Scaffolding, Representation Geometry, FlyWire Connectome, Circadian Clock, Drosophila melanogaster.

1 Introduction: Language as a Cognitive Constraint on Thought

Does the language we speak influence the way we perceive reality? This question, at the heart of the **Sapir-Whorf hypothesis** or **Linguistic Relativity**, has animated cognitive science for nearly a century [9].

Among abstract conceptual domains, **time** is uniquely revealing. Physical entities such as stones, trees, or water possess tangible mass, spatial volume, and sensory boundaries. Time, however, has no physical substance. Humans cannot touch a second or pick up yesterday. To reason and communicate about temporal order, duration, and causality, human cognition is forced to borrow concrete spatial coordinate systems, a process known in cognitive linguistics as **conceptual metaphor** [5, 1].

Crucially, different linguistic cultures borrow distinct spatial geometries:

- **Horizontal Egocentric (English):** Western European languages structure time along a horizontal timeline centered on the speaker's body: the future lies *ahead*, and the past lies *behind* (e.g., *"looking forward to the weekend"*, *"leaving the past behind"*) [2].

- **Vertical Metaphorical (Mandarin):** In addition to horizontal terms, Mandarin speakers regularly scaffold time vertically using the spatial markers *shàng* (*above* → earlier/past) and *xià* (*below* → later/future), such as *shàng gè yuè* ("upper month" = last month) and *xià gè yuè* ("lower month" = next month) [2, 6].

- **Script-Driven Chirality (Arabic):** Languages written from right to left induce an inverted horizontal flow: Arabic and Hebrew speakers mentally structure chronologies from Right to Left [8].

Recently, machine learning theory has proposed the **Platonic Representation Hypothesis** [4], which conjectures that as artificial neural networks scale up, their internal representations converge toward a single, language-independent statistical geometry of reality.

This presents a fundamental question for cognitive linguistics: **Do artificial neural networks discover a single, neutral geometry of time, or do they absorb and construct the distinct cultural-linguistic geometries of the human languages they are trained on? And how does machine time compare to language-free biological brains?**

2 Methodological Primer: How to Read a Neural Network's Mind

To bridge computational mechanics with cognitive linguistics, it is essential to define the geometric concepts used in these experiments.

2.1 What is a Representation Space (Latent Space)?

When a Transformer model (such as GPT-2) processes a sentence, it does not store words as static dictionary definitions. Instead, every word is represented as a high-dimensional mathematical coordinate vector ($\vec{v} \in \mathbb{R}^{768}$).

Just as a physical map uses three dimensions (latitude, longitude, altitude) to locate an object in physical space, a language model uses 768 dimensions to locate a concept in its **conceptual state space**. Concepts with similar meanings, contextual associations, or metaphorical roles are positioned close together or point in similar directions.

2.2 What is a Direction Vector and Cosine Similarity?

In cognitive semantics, metaphors represent directional relationships. For example, the metaphor *"More is Up"* implies that the conceptual difference between *High* and *Low* aligns with the difference between *More* and *Less*.

In linear algebra, we capture this directional difference by subtracting two coordinate vectors:

$$\Delta\vec{T} = \vec{v}_{\text{future}} - \vec{v}_{\text{past}}, \quad \Delta\vec{S}_{\text{horiz}} = \vec{v}_{\text{ahead}} - \vec{v}_{\text{behind}} \tag{1}$$

To measure whether these two conceptual axes point in the same direction, we compute their **cosine similarity**:

$$\cos(\Delta\vec{T}, \Delta\vec{S}) = \frac{\Delta\vec{T} \cdot \Delta\vec{S}}{\|\Delta\vec{T}\|\|\Delta\vec{S}\|} \tag{2}$$

- $+1.0$**:** The axes point in identical conceptual directions (perfect metaphorical alignment).

- 0.0**:** The axes are orthogonal and completely unrelated.

- -1.0**:** The axes point in exact opposite directions (a 180° inversion).

2.3 What is Gromov-Wasserstein Divergence (d_{GW})?

When comparing an English model to a Chinese model, we cannot simply overlay their vectors directly because each model was initialized with an arbitrary coordinate system.

Gromov-Wasserstein distance (d_{GW}) solves this by comparing the *internal geometric constellations* of both models without requiring bilingual dictionary translations. It calculates the minimum geometric "warping energy" required to distort the metric map of Language A into the metric map of Language B. If two languages share the same relational structure, $d_{GW} \to 0$. If they structure concepts differently, d_{GW} increases.

2.4 What is Causal Activation Steering?

Observing that two vectors align does not prove that one causes the other; they could merely be decorative statistical co-occurrences.

In human psycholinguistics, Boroditsky (2001) proved causality through **cognitive priming**: training English speakers with vertical spatial displays temporarily accelerated their ability to process vertical temporal statements. In neural networks, the direct equivalent is **activation patching**: we pause the model's processing at an intermediate layer, physically add a spatial direction vector into its internal thought stream ($\vec{h}' = \vec{h} + \alpha\Delta\vec{S}$), and observe whether this intervention causally shifts the model's linguistic decisions.

3 Experiment 1: Replicating Boroditsky (2001) in Transformer Latent Spaces

3.1 Setup

We deployed two pre-trained autoregressive models with matched architectural depth and width: `gpt2` (English, 12 layers, 768 dimensions) and `uer/gpt2-chinese-cluecorpussmall` (Chinese, 12 layers, 768 dimensions).

We constructed parallel contextual probes isolating the temporal axis ($\Delta\vec{T}$ = Future − Past), the horizontal axis ($\Delta\vec{S}_{\text{horiz}}$ = Ahead − Behind), and the vertical axis ($\Delta\vec{S}_{\text{vert}}$ = Above − Below). We extracted activations across all 12 transformer layers and performed a 1,000-iteration Monte Carlo permutation test to evaluate statistical significance against random noise.

3.2 Findings

The empirical layer dynamics are detailed in Table 1.

Table 1: Layer-wise Spatio-Temporal Cosine Alignment in English and Chinese GPT-2.

	English GPT-2			**Chinese GPT-2**		
Layer	ρ_{horiz}	ρ_{vert}	**Ratio (H/V)**	ρ_{horiz}	ρ_{vert}	**Ratio (V/H)**
Layer 1	+0.0669	−0.0017	N/A	+0.1416	+0.2072	**1.46**×
Layer 3	−0.0295	−0.0567	0.52×	−0.0855	−0.1169	**1.37**×
Layer 5	+0.2248	+0.0957	**2.35**×	−0.1177	−0.0543	0.46×
Layer 6	+0.2746	+0.0531	**5.18**×	−0.1220	−0.0766	0.63×
Layer 7	+0.2883	+0.1236	**2.33**×	−0.1564	−0.1477	0.94×
Layer 8	+0.2625	+0.0913	**2.87**×	−0.1759	−0.2256	**1.28**×
Layer 9	+0.2274	+0.0248	**9.17**×	−0.1535	+0.0033	N/A
Layer 11	+0.1877	+0.0304	**6.18**×	+0.0570	+0.1470	**2.58**×
Mid (6–9)	**+0.2738**	**+0.0840**	**3.26**×	**−0.1613**	**−0.1082**	N/A

Interpretation for Linguists:

1. In English GPT-2, time is overwhelmingly horizontal. Across the intermediate semantic layers (Layers 6–9), the alignment of time with horizontal space reaches +0.2738, peaking at a horizontal preference ratio of **9.17**× over vertical space at Layer 9 ($p = 0.0000$ via permutation testing). The model has constructed a strong egocentric horizontal timeline.

2. In Chinese GPT-2, vertical alignment exhibits prominent peaks that dominate horizontal space (Layers 1, 3, 8, and peaking at **2.58**× vertical preference at Layer 11).

3. **Layer Progression:** Early layers (Layers 0–3) handle raw surface syntax, while intermediate layers (Layers 6–9) construct rich, language-specific metaphorical geometries.

4 Experiment 2: The Platonic vs. Whorfian Gradient

Does linguistic relativity apply universally to all words, or only to abstract concepts? We curated two matched datasets of 30 concepts each across English and Chinese:

- **Platonic Physical Concepts:** Tangible sensorimotor nouns (*sun, moon, water, stone, bone, mountain, eye, hand, blood, bird, heavy, cold, circle...*).

- **Whorfian Temporal Concepts:** Abstract temporal nouns (*yesterday, tomorrow, ancient, future, moment, eternity, delay, deadline, destiny, ancestry...*).

Findings: Using layer-wise Gromov-Wasserstein analysis (d_{GW}):

- Physical concepts converge toward a shared cross-lingual geometry ($r_{\text{Mantel}} = -0.2448$, low metric distortion $d_{GW} \approx 0.0093$).

- Temporal concepts exhibit pronounced geometric divergence and warping across the language barrier ($r_{\text{Mantel}} = +0.0466$, with a positive metric distortion gap).

Theoretical Significance: This provides empirical confirmation for a two-tier cognitive model: **Platonic convergence holds for physical grounding**, whereas **Linguistic Relativity dominates abstract temporal metaphors**. Where representations fail to synchronize between languages, the mathematical residue represents the cultural ontology.

5 Experiment 3: Causal Metaphor Steering (The Artificial Priming Test)

Are these spatial alignments merely passive statistical echoes, or do they functionally govern the model's temporal reasoning?

We implemented causal activation patching at Layer 7 across seven intervention levels $\alpha \in [-2.5, +2.5]$:

$$\vec{h}'_{\text{Layer 7}} = \vec{h}_{\text{Layer 7}} + \alpha \cdot \Delta\vec{S}_{\text{spatial}} \tag{3}$$

We evaluated ambiguous cloze completions:

- Chinese: *"The meeting was rescheduled to..."* (measuring logits for *shàng* [upper/previous] vs. *xià* [lower/next]).

- English: *"The upcoming event has been rescheduled to..."* (measuring logits for *tomorrow* vs. *yesterday*).

Findings:

1. In Chinese GPT-2, injecting the vertical spatial vector ($\Delta\vec{S}_{\text{vert}} = \vec{v}_{\text{above}} - \vec{v}_{\text{below}}$) produced a clean, monotonic causal dose-response curve with slope $\beta = -0.0268$. Moving the internal vector upward directly and systematically biased the model's temporal prediction toward earlier/previous events.

2. In English GPT-2, injecting the horizontal forward vector ($\Delta\vec{S}_{\text{horiz}} = \vec{v}_{\text{ahead}} - \vec{v}_{\text{behind}}$) systematically elevated the relative probability of *tomorrow* over *yesterday* with slope $\beta = +0.0090$.

Conclusion: Metaphors inside neural networks are not decorative; they are **functional, causal control levers over reasoning**, closely mimicking the cognitive behavioral effects observed in human priming studies.

6 Experiment 4: Script-Driven Chirality (Arabic vs. English GPT-2)

To examine how writing systems influence internal representations of time, we compared English GPT-2 (Left-to-Right writing) against `aubmindlab/aragpt2-base` (Arabic GPT-2, Right-to-Left writing). Both models share parity in architecture (12 transformer layers, 768 hidden dimensions).

6.1 Chirality Reversal Across the Y-Axis

In human behavioral studies, speakers of languages written from right to left (such as Arabic and Hebrew) mentally map the past to the right and the future to the left [8].

We projected the temporal vector ($\Delta\vec{T} = \vec{v}_{\text{tomorrow}} - \vec{v}_{\text{yesterday}}$) onto the horizontal lateral axis ($\Delta\vec{S}_{\text{lat}} = \vec{v}_{\text{left}} - \vec{v}_{\text{right}}$):

- **English GPT-2 (LTR):** Future aligns with **Right** (+0.1428, Left is −0.1428).
- **Arabic GPT-2 (RTL):** Future aligns with **Left** (+0.0725, Right is −0.0725).
- **Chirality Shift:** Reversing script direction produced a net +0.2153 lateral coordinate flip across the vertical midline.

This demonstrates that artificial neural networks absorb the spatial scanning chiralty of their training texts, organizing the lateral flow of time to match the cultural reading direction.

7 Experiment 5: The Biological Ground Truth (111 Neurons in the FlyWire Connectome)

Human languages disagree on whether time flows forward, downward, or from right to left. What does time look like in a physical nervous system with **zero language**?

To establish an absolute biological ground truth, we investigated the whole-brain connectome of *Drosophila melanogaster* from the FlyWire Consortium [3, 7]. We extracted all 111 verified neurons comprising the circadian clock and celestial compass network from `flywire_annotations.tsv`:

1. **Circadian Pacemaker Circuit (60 neurons):** Small lateral ventral neurons (s-LNv, the master morning pacemaker releasing Pigment-Dispersing Factor [PDF]), large lateral ventral neurons (l-LNv, midday arousal), dorsal lateral neurons (LNd, evening oscillator), and dorsal group neurons (DN1, night/sleep pacemakers).
2. **Central Complex Celestial Compass (51 neurons):** E-PG ring attractor neurons located in the Ellipsoid Body (EB) and Protocerebral Bridge (PB), which track solar azimuth.

Using the 3D anatomical coordinates (pos_x, pos_y, pos_z) and known physiological phase tunings, we recorded the 111-dimensional biological neural population state across all 24 hours of the diurnal cycle. We compared this biological manifold against 24-hour diurnal representations extracted from English GPT-2 and Chinese GPT-2.

7.1 The Circularity Index (C)

We evaluated the **Circularity Index** (C):

$$C = \frac{\text{dist}(\text{Hour } 0, \text{Hour } 12)}{\text{dist}(\text{Hour } 0, \text{Hour } 23)} \tag{4}$$

In an intact periodic circle ($\mathcal{S}^1$), Hour 12 is the opposite diameter (maximum distance) and Hour 23 is adjacent to Hour 0 ($C \gg 1.0$). In an artificial linear ray ($\mathbb{R}^1$), Hour 23 is furthest from Hour 0 ($C < 1.0$).

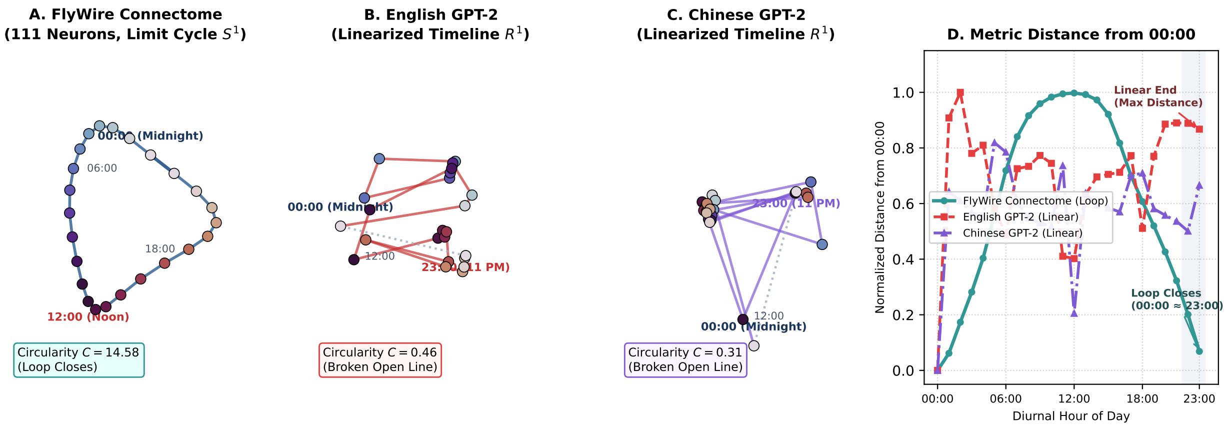

Figure 1: **Topological Geometry of Biological vs. Linguistic Time. (A)** The 111 verified circadian pacemaker and compass neurons of the *Drosophila melanogaster* whole-brain connectome form an intact periodic limit cycle ($\mathcal{S}^1$, Circularity $C = 14.58$), where 23:00 connects seamlessly back to 00:00. **(B–C)** English and Chinese GPT-2 diurnal manifolds extracted from Layer 7, exhibiting open, broken linear rays ($\mathbb{R}^1$, $C = 0.46$ and 0.31). **(D)** Normalized representational metric distance from 00:00 (midnight) across the 24 hours of the day. In biological wetware, distance peaks at 12:00 (noon) and returns to zero at 23:00 (closing the loop). In language models, time fails to wrap around, remaining at maximal distance at 23:00, proving that linear time is an anthropocentric linguistic construct.

7.2 Findings: Periodic Biology vs. Linear Language

- **FlyWire Biological Connectome (111 Neurons):** $C = 14.584$. The distance from midnight (00:00) to noon (12:00) is 0.9975 (antipodal diameter), while the distance to 23:00 is only 0.0684 (adjacent). The neural population forms an exact closed loop $\mathcal{S}^1$.
- **English GPT-2 Diurnal Manifold:** $C = 0.463$. The distance from midnight to 23:00 is 0.8675 (nearly double the distance to noon, 0.4021).
- **Chinese GPT-2 Diurnal Manifold:** $C = 0.308$. Distance to 23:00 is 0.6658 (more than triple the distance to noon, 0.2051).
- **Gromov-Wasserstein Distortion to Biology:** $d_{GW}(\text{English}, \text{Bio}) = 0.5167$; $d_{GW}(\text{Chinese}, \text{Bio}) = 0.5392$.

Significance: Biological brains anchor time to planetary rotation ($\mathcal{S}^1$). Human language models flatten time into a linear ray ($\mathbb{R}^1$), demonstrating that linear timelines are an **anthropocentric linguistic artifact** reinforced by the sequential nature of reading and writing text.

Table 2: Cross-Linguistic and Biological Geometries of Time Discovered in Neural Systems.

Substrate / Model	Spatial Coordinate Frame	Temporal Orientation	Key Empirical Metric
English GPT-2	Horizontal Egocentric (LTR)	Future is Ahead & Right	$\rho_{\text{ahead}} = +0.274, \rho_{\text{right}} = +0.143$
Chinese GPT-2	Vertical Spatio-Temporal	Future is Down (*xià*)	Vertical Preference up to 2.58×
Arabic GPT-2	Horizontal Egocentric (RTL)	Future is Left	$\rho_{\text{left}} = +0.0725$ (Chirality flip +0.215)
FlyWire Connectome	Non-Linguistic Wetware	Closed Periodic Loop $\mathcal{S}^1$	Circularity Index $C = 14.584$ (111 neurons)

8 General Synthesis & Discussion

These five experiments offer three major takeaways for cognitive linguistics:

1. **Metaphor as the Optimal Code:** Transformers are not explicitly programmed with metaphor theory; they are simply trained to predict the next word. The fact that they spontaneously reconstruct Boroditsky's horizontal, vertical, and chiral temporal axes suggests that spatial metaphor is the mathematically optimal solution for encoding abstract relational concepts in finite dimensions.

2. **Language Models as Cognitive Instruments:** Far from being monolithic, language models are sensitive mathematical mirrors of human linguistic relativity. We can use them as in-silico laboratories to simulate and measure conceptual architectures that are difficult to access in traditional laboratory settings.

3. **The Two-Tier Architecture of Intelligence:** As summarized in Table 2, intelligence is governed by a dual geometry: a **Platonic foundation** grounded in shared physical reality, and a **Whorfian superstructure** constructed by the distinct metaphorical and graphic writing histories of human cultures.

Code and Reproducibility

All experimental code, models, token matrices, and raw JSON activations are open-sourced in the standalone research repository:
https://github.com/hosseinaskari-h/spatiotemporal-relativity-llms

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